ALGEBRA 1 · SOLVING LINEAR EQUATIONS

Equation Balance Scale

Explore & Reflect

Name _____ Date _____ Period _____

Open **Equation Balance Scale** on Mr. K's Classroom: Algebra 1 → Solving Linear Equations → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. On the scale for 2x + 3 = 11, what will happen if you subtract 3 from the left side only?

2. To solve 2x + 3 = 11, which operation would you do first? Why?

2 Explore use the tool and record what you see

Solve on the scale

Use only operations on both sides. List your steps, then check your answer in the original equation.

Equation	Your steps	Solution	Check
A: 2x + 3 = 11			
B: 3x − 5 = 10			
C: 5x + 2 = 2x + 14			
D: 6x − 4 = 2x + 8			

When the x-terms cancel

In equations E and F, subtract the x-terms from both sides.

Equation	What is left	True or false?	Number of solutions
E: 2x + 10 = 2x + 10			
F: 3x + 4 = 3x − 2			

3 Write the rule fill in each blank from what you observed

1. To keep an equation balanced, whatever you do to one side you must also do to _____.

2. To undo addition, _____. To undo multiplication, _____.

3. If the x-terms cancel and what is left is true (like 10 = 10), the equation has _____.

4. If the x-terms cancel and what is left is false (like 4 = −2), the equation has _____.

4 Check no tool: show your work

1. Solve $4x - 7 = 13$.

2. Solve $7x + 3 = 4x + 15$.

3. Solve $\frac{x}{3} + 2 = 6$.

4. How many solutions does $2(x + 5) = 2x + 10$ have?

5. How many solutions does $5x - 1 = 5x + 6$ have?

5 Reflect

Why do you have to do the same thing to both sides of an equation?